

فیزیولوژی

1- Given your knowledge regarding the response of the pulmonary vasculature to oxygen content, where is the highest amount of blood flow through a normally oxygenated lung in a normal weight adult male that is standing upright?

- ✓ To the bottom of the lung
- To the midzone of the lung
- To the top of the lung
- There is no gradient across the lung

2- Which one of the following center is thought to control of the respiratory rate?

- Apneustic center
- Ventral respiratory group of medullary neurons
- Dorsal respiratory group of medullary neurons
- ✓ Pneumotaxic center

3- What is the effect of an increase in the CO₂ of blood?

- Decreases dissociation of O₂ from oxyhemoglobin
- Increases the amount of oxyhemoglobin
- ✓ Decreases the affinity of oxyhemoglobin for oxygen
- Shifts the oxyhemoglobin dissociation curve to the left

4- What is the the initial pressure difference (mmHg) that causes O2 to diffuse into the pulmonary capillary?

17

21

11

✓ 64

5- Which one of the following list includes only structures that compose the anatomic dead-space?

Pharynx, bronchi, alveoli, larynx

Respiratory bronchioles, alveolar ducts, alveoli

Respiratory bronchioles, alveoli, trachea

✓ Trachea, bronchi, terminal bronchioles

6- What is the effect of decreased alveolar oxygen on local alveolar blood flow?

Increases

It first increases and then decreases

✓ Decreases

It first decreases and then increases

7- Which one of the following occurs during normal inspiration?

Diaphragm contracts, pleural pressure increases, alveolar pressure decreases

Diaphragm relaxes, external intercostals contract, pleural pressure increases

External and internal intercostals contract, pleural and alveolar pressure increase

✓ Diaphragm contracts, pleural and alveolar pressures decrease

8- Which one of the following is NOT a major function of the Upper Track?

- Warm the air
- Humidify the air
- Filter the air
- ✓ Gas diffusion

9- Which zone(s) is (are) seen in a healthy subject during exercise?

- Zone-II
- ✓ Zone-III
- Zone-I and II
- Zones II and III

10- Which one of the following gas has more "diffusing capacity" than others in resting?

- Nitrogen
- Carbon monoxide
- Oxygen
- ✓ Carbon dioxide

11- In response to hypoxia, which neurotransmitter is NOT released by "Glomus cells" in the carotid bodies?

- ✓ Histamine
- Dopamine
- ATP
- Acetylcholine

12- If the radius of alveoli increases two times and the surface tension remains constant, what will change the alveolar collapse pressure?

- It will double
- It will not change because the surface tension remains constant.
- It will increase to four times
- ✓ It will be halved ($1/2$)

13- Which one of the following is TRUE about bronchial circulation?

- Has vasoconstriction in response to hypoxia
- ✓ Supplies blood flow to conducting zone
- Provides pulmonary capillary perfusion for gas exchange
- Is a part of the the low-pressure circulation

14- What is the minute respiratory volume (L/min); with a tidal volume of 0.8 L and respiratory rate of 15 breaths/min?

- ✓ 12
- 4.2
- 8.5
- 6

15- What is the cause of "physiological shunt" occurrence?

- ✓ Decreasing of ventilation/perfusion ratio (V_A/Q) to zero
- Mixing of pulmonary vein blood with bronchial vein blood
- Increasing of V_A/Q ratio more than normal amount
- Decreasing the cardiac output to half

16- What is the chloride shift in RBC of venous blood?

chloride ion goes out of the cell and H^+ goes into the cell

HCO_3^- goes into the cell and chloride ion goes out of the cell

- ✓ chloride ion goes into the cell and HCO_3^- goes out of the cell
- H^+ goes into the cell and chloride ion goes out of the cell

17- What is the Recoil pressure?

- ✓ Alveolar pressure minus plural pressure
- Plural pressure minus alveolar pressure
- Atmosphere pressure minus plural pressure
- Alveolar pressure minus atmosphere pressure

18- If a patient had a progressive lung disease that required an ever increasing pressure to fill the same volume of lung, how would the lung's compliance be affected?

- It would increase it.
- It would stay the same
- ✓ It would decrease it.
- These variables do not affect lung compliance.

19- In which one the following, the diffusion rate of gases is decreased between two sides of respiratory membrane?

- Increasing the surface area
- ✓ Removal of part of the lungs
- Increasing the gas pressure difference between the two sides of the membrane
- Decreasing the thickness of the membrane

20- Which one of the following has afferent nerve fibers in the vagus nerve?

- ✓ Aortic body
- Chemosensitive area in medulla
- Carotid body
- Hypoxia- induced vasoconstriction

21- Which one of the following is TRUE about “the duration of expiration”?

- Is less than duration of inspiration during normal and quite respiration
- Is affected by ventral respiratory group (VRG) during normal respiration
- Increases by strong activation of pneumotaxic center
- ✓ Decreases when respiratory rate increases

22- What is “the maximum amount of air that can be expired after a maximum inspiration”?

- Total lung volume
- Tidal volume
- ✓ Vital capacity
- Expiratory reserve volume

بافت

23- In which of the following is observed Elastic Fiber In Histology Stractuer ?

Alveolar Duct

Vestibule Zone



Larynex

Respiratory Zone

24- In which of the following is observed Stratified Squamous Tissue?

Olfactory Zone

Respiratory Zone

Pharynex



Vestibule Zone

25- Which of the following is observed in structure Histology In Respiratory Zone?

Dense Connective Tissue

Stratified Squamous Epithelial Tissue



Goblet Cell

Dense Connective Tissue + Erectile Tissue

26- What Is The Name Cell Secreted Surfactant In Bronchios?

- ✓ Celara
- Macrophage
- Pnomocyte III
- Simpel Cells

27- Which part of the lung tissue has Nomocyte II?

- Terminal Bronchiole
- ✓ Alveolar
- Bronchiole
- Bronchus

آناتومی**28- All of the following muscles are paralyzed by cutting the recurrent laryngeal nerve except?**

- Thyroepiglottic
- Lat Crico- arytenoid
- ✓ Cricothyroid
- Posterior crico- arytenoid

29- All of the following bones participates in the formation of the lateral wall of the nose except?

- Lacrima
- Palatine
- Maxilla
- ✓ Vomer

30- Which of the following nerve pass through incisive canal ?

- Anterior ethmoidal
- ✓ Nasopalatine
- Greater palatine
- Posterior ethmoidal

31- Which of the following arteries passes through the foramen in roof of nasal cavity?

- Sphenopalatin artery
- Superior labial artery
- Greater palatine artery
- ✓ Anterior ethmoidal artery

32- Which bone is involved in both the roof and the medial surface of the nasal cavity?

- Ethmoid
- ✓ Palatine
- Sphenoid
- Inferior concha nasalis

33- Which of the following paranasal sinuses open to the apex of bulla ethmoidal ?

- ✓ Middel ethmoidal sinus
- Anterior ethmoidal sinus
- Frontal
- Maxillary

34- All of the following structures are seen in inlet of larynx except?

- Arythnoid cartilage
- Ary-epiglottic fold
- Epiglot cartilage
- ✓ Vestibular fold

35- What is the name of the upper and free side of the quadrangular membrane?

- Median thyrohyoid ligament
- ✓ Ary -epiglottic fold
- Vocal ligament
- Vesibular ligament

36- The medial surface of the right lung is adjacent to all the following structures except

- ✓ Arch of aorta
- Vena cava superior
- Azygos vein
- Vena cava inferior

37- Which of the following muscles is involved in pulling thyroid cartilage downward and forward ?

...

Transverse arytenoid



Cricothyroid

Oblique arytenoid

Posterior crico- arytenoid

38- Which of the following is the highest structure in the right lung umbilicus?

Pulmonary vein

Main bronchus



Superior upper lobe bronchus

Pulmonary artery

جنین شناسی

39- In which period of lungs development alveolar duct appear



Canalicular phase

Alveolar phase

Saccular phase

Pseudoglandular phase

40- In which time of development the second epithelial alveolar cells appear?

4th month

5th month

✓ 6th month

3th month